

Tide retrospective: *first derivative of the cumulant generating function*

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Abstract

The first rung of the connected-cumulant ladder for the anharmonic perturbed partition. With $Z(0) > 0$ and the partition's first h -derivative in hand, `HasDerivAt.log` gives the cumulant generating function's first derivative

$$K'(0) = \frac{Z'(0)}{Z(0)} = \langle -(tx) \rangle_0,$$

the unperturbed Gibbs mean of the perturbation observable $-(tx)$. The new file¹ builds clean, no `sorry/axiom/native_decide`; first try.

1 Setting

This is the fifth tide of a single-day anharmonic arc on the laplace seabed (FDT capstone; the iterated-derivative step and its general- h strengthening; moment-normalisation). Those gave the partition $Z = \text{weightedPartition} \dots 0$, its derivatives $Z^{(n)}(0) = \int (-tx)^n e^{-tL}$, and the reading $Z^{(n)}(0)/Z(0) = \langle (-tx)^n \rangle_0$. The natural next object is the *cumulant* generating function $K = \log Z$, whose derivatives at 0 are the connected cumulants. This tide lands $K'(0)$ — the lowest rung, and the foundation for $\kappa_2, \kappa_3, \kappa_4$. It was the lowest-risk way to open the cumulant direction at the end of a long session: a single `HasDerivAt.log` application.

2 The thing formalised

For the anharmonic potential, $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$:

$$\text{HasDerivAt } (h \mapsto \log Z(h)) \left(\frac{Z'(0)}{Z(0)} \right) 0, \quad \text{deriv } (h \mapsto \log Z(h))(0) = \langle -(tx) \rangle_0,$$

the second equality identifying $K'(0)$ with the unperturbed Gibbs mean.

```
theorem logPartition_hasDerivAt ... :
  HasDerivAt (fun h => Real.log (weightedPartition lam alpha gamma t 0 h))
    (weightedPartition lam alpha gamma t 1 0
      / weightedPartition lam alpha gamma t 0 0) 0
```

¹AnharmonicLogPartitionDeriv.lean.

3 Proof strategy

Three short steps. (i) $Z(0) = \text{weightedPartition} \dots 00 = \int e^{-tL}$ by `pow_zero/one_mul` and a ring on the exponent, hence > 0 by `anharmonic_partition_pos`. (ii) The previous tide's `weightedPartition_hasDerivAt` at $n = 0$, $h_0 = 0$ gives `HasDerivAt Z (Z'(0)) 0`; feeding it and $Z(0) \neq 0$ to `HasDerivAt.log` yields the CGF derivative $Z'(0)/Z(0)$. (iii) The identification with the Gibbs mean is moment-normalisation at $n = 1$ (rewriting $Z'(0) = \text{iteratedDeriv 1 Z 0}$) followed by `pow_one` to turn $-(tx)^1$ into $-(tx)$.

4 Roadblocks and resolutions

None — it compiled on the first build. The one thing to note is the defeq $0 + 1 = 1$: `weightedPartition_hasDerivAt 0` produces the derivative `weightedPartition ... (0+1) 0`, which exact happily unifies with the stated `weightedPartition ... 1 0`.

5 What was Mathlib, what was new

Mathlib. `HasDerivAt.log`, `HasDerivAt.deriv`, `pow_zero`, `pow_one`, `one_mul`. **Seabed (reused).** `weightedPartition_hasDerivAt`, `anharmonic_partition_pos`, and the moment-normalisation and `iteratedDeriv` identities — all from this session's arc. **New.** The $Z(0)$ positivity helper and the two CGF-derivative theorems. About 70 lines.

6 Lessons

- **Build the generating function one derivative at a time.** `HasDerivAt.log` on the partition's first derivative is a clean, low-risk way to open the cumulant ladder; the higher cumulants are then iterated `HasDerivAt.div/HasDerivAt` compositions on top.
- **A foundation tide need not be large to be load-bearing.** $K'(0)$ is short, but every connected cumulant the arc will want is a derivative of this object.

7 Follow-ups

- **Second cumulant** $\kappa_2 = K''(0)$. The connected variance $\langle u^2 \rangle_0 - \langle u \rangle_0^2$, via `HasDerivAt.div` on Z'/Z (using the general- h step so Z' is differentiable on a neighbourhood). Then κ_3, κ_4 — the fourth-cumulant / flow-equation identity (survey v4 #2).
- **Promote** `amgm_t_abs_x` to `lean-common`.